



FIRST SEMESTER 2026-27

COURSE HANDOUT (PART II)

In addition to part-I (General Handout for all courses) printed on page 1 of the Time Table booklet, this portion gives further specific details regarding the course.

Course No. : BIO G512
Course Title : Molecular Mechanism of Gene Expression
Instructor- in Charge : S. K. VERMA
Sudeshna Mukherjee

1. Course Description:

Prokaryotic and Eukaryotic genomes and their topology; DNA protein interaction, RNA transcription and transcriptional control, Translation, RNA processing, Post transcriptional control and other mechanisms of gene expression

2. Scope & Objective:

This course is designed to impart knowledge of molecular genetics, which is one of the fundamental requirements for the biotechnology. This course is designed to make the student understand the various mechanisms, which regulate the biological processes at genetic level including transcriptional and post transcriptional regulation of gene.

3. Text Books:

Lewin's Genes XI: Jocelyn Krebs, Elliott Goldstein and Stephen Kilpatrick; (11th Ed), Jones & Bartlett Learning Publication, 2014

Reference Book:

Molecular Biology of Gene: Watson, Baker, Bell, Gann, Lavine & Losick (5th Ed)
Pearson Education





4. Course Plan:

Module Number	Lecture session	Reference	Learning Outcome
1. Introduction to genome	L 1-5. Genes are DNA, Different methods to map a gene in prokaryote and eukaryotes. Fine structure of Gene L 6-8. Alternative states of chromatin, chromatin remodeling, Alteration of nucleosome organization at promoter. Histone acetylation & modification, Histone methylation & phosphorylation	Lecture Notes & Chap 10	Content and organization of genome
2. Transcription	L 9-12 Prokaryotic Transcription Three stage of Transcription, RNA Polymerase structure, Promoter recognition, and efficiencies, Sigma factor, alternate sigma factors and its binding with DNA, termination of transcription; Intrinsic and rho-dependent L 13-16 Eukaryotic Transcription: RNA polymerase I, II, III & Types of Promoters, Enhancers, Transcription factors, Assembly of basal apparatus, Insulation block	Chap 19 & Chap 20	Transcription process in microbes & Higher organism
3. RNA Splicing and Processing	L 17-22. 5' Cap, Nuclear Splicing, Splice junctions, lariat formation and spliceosome, Splicing of group I & II introns, Splicing & gene Expression,	Chap.21 & 23	Post Transcriptional Mechanisms



	Alternative splicing, Trans-splicing, 3' End Processing, tRNA & rRNA splicing, ribozyme, Ribonuclease activities and RNA editing, Guide RNA		
4. Translation and Genetic code	L 23-27 Stages in protein synthesis, Initiation of translation in bacteria, Elongation, Termination Genetic code: Structure and wobbling in codons, anti-codones, Mitochondrial and archetypal code, mutated anticodones, Accuracy of translation	Chap.24-25	Translation in Prokaryotes and Eukaryotes
5. Gene Regulation	L 27-31. <u>Operon Circuits</u> Structural gene clusters, Operator and regulator gene, Repressor protein structure, Specificity of protein-DNA interaction, binding of repressor with operator and inducer. Allosteric changes in repressor. <i>lac</i> and <i>trp</i> operon. L 32-36 <u>Transcription Regulation in Eukaryotes</u> Interaction of activators with basal apparatus, Methylation of histones and DNA, DNA binding domains, Zinc fingers, steroid receptors, leucine zipper, Histone acetylation and modification, Yeast <i>GAL</i> genes	Chap.26 & Chap 28	Gene Regulation in Prokaryotes & Eukaryotes
6. Phage Strategies	L 37-40 Control of lytic cascades in T4, T7 and lambda phage, Maintenance of lysogeny, Repressor synthesis	Chap 27	Regulation of Lysogeny and lytic cycle



	and binding, Balance between lysogeny and lytic cycle		
7. Epigenetic Effects	L 41-42. Histone methylation and phosphorylation, Epigenetic effects	Chap 29	Different phenotypic consequences without any change in the sequence of DNA

5. Examination Scheme:

EC No	Evaluation Component	Duration	Date	Weightage (%)	Remarks
1	Mid- Sem Test	90 min		25	CB
3.	Quiz (multiple)	-		10	CB
4.	Seminar +Project (Multiple)	-	To be announced	30	OB
5.	Comprehensive Examination	3Hrs.		35	OB + CB

6. Chamber Consultation Hour: To be announced

7. Grading Policy: Award of grades would be guided by the histogram of marks and course average. **If a student obtains below 30% of median score in the course, the student will not be awarded a valid grade and the case will be reported as “NC”.**

8. Notices: Notices will be displayed on Nalanda.

9. Make-up Policy: Prior Permission has to be obtained from the Instructor-In-Charge, no make-up will be granted for surprise evaluation component.

Instructor-in-Charge
BIO G512